SOC Alert Lifecycle Overview

Objective: To learn how alert flows inside a SOC, the Tiers of SOC and also understand the alert lifecycle.

1. What Is a SOC?

A SOC is a team of cybersecurity professionals that is responsible for monitoring, detecting, analyzing and responding to cybersecurity threats in an organization's IT infrastructure. The SOC operates 24/7 to ensure threats are identified early and handled effectively before business operations are impacted.

2. SOC Team Structure (Tier 1, Tier 2, Tier 3)

- Tier 1 - SOC Analyst(Monitoring & Triage)
  Monitors security alert from security tools like SIEM, perform initial analysis of alert, verifies if its a false positive or a true positive alert, and escalates verified alert to tier 2.

- Tier 2 - SOC Analyst (Investigation & Response)
  Performs deep investigation of alerts, Determines attack scope and impact, and initiates the containment and remediation actions.

- Tier 3 - SOC Engineer / Threat hunter
  Handles advanced persistent threats, Performs malware analysis and threat hunting, Improves detection rules and SOC tools, and also helps in incident response and forensic analysis.

3. What is an alert?

An alert is a notification generated from a security tool when a suspicious or abnormal behaviour is detected.
Common alerts triggers are;

- Multiple failed login attempts
- unusual outbound traffic to suspicious IP address
- Privilege escalation attempts
- Malware detected by an antivirus or EDR

Alerts are signals, not confirmation of an attack.

4. Difference between alert and Incident.

Alert: A warning that a suspicious activity may be happening.

Incident: A confirmed security event that violates security policy.

5. **False positive vs True Positive**

False Positive: This is an alert that is generated by a legitimate activity. No real security threat happens.
E.g : employee logins from a new location

True Positive: This is an alert that correctly identifies a malicious activity.
E.g : A phishing attack that steals credentials information.

6. **SOC Alert lifecycle**
   - Detect: Security tools(SIEM, EDR, IDS) generate alerts.
   - Triage: Tier 1 analyst reviews alert details , checks severity, source and context. Determines if the alert is false or suspicious.
   - Investigate: Tier 2 analyst performs deeper analysis, review logs, endpoints and network traffic. Confirms if it's a true security threat.
   - Escalate/Close: Escalate if it has been confirmed as a security incident. If the alert is false positive or low risk, document and close.

7. **What SOC Analysts Do With Alerts**
   - Validate alerts using logs and tools
   - Investigate suspicious behaviour
   - classify alert base on severity and impact
   - Escalate confirmed threat
   - Document findings
   - Improve detection rules based on lesson learned

Summary

A SOC ensures security threats are detected early, investigated properly, and handled efficiently. This exercise strengthened my understanding of SOC operations, alert handling, and the difference between alerts and incidents in real-world security environments.